Customer Churn Prediction Final Report

Content

List of Figures

List of Tables

1. Understanding the Business Problem

1. Defining the Problem Statement

AlphaCom is currently facing a significant increase in customer churn, which is negatively impacting both revenue and brand loyalty. Despite offering competitive services, the company is unable to effectively retain customers due to the lack of actionable insights into churn behavior.

Customer churn is not driven by a single factor but rather a complex interaction of multiple variables such as Service usage patterns, Billing preferences, Contract types, Customer demographics

The absence of a predictive and data driven approach has resulted in reactive decision making, where retention strategies are implemented only after customers have already churned.

Problem Statement:

To identify customers who are likely to churn and understand the key factors influencing their decision, enabling proactive and targeted retention strategies.

2. Need of the Study / Project

This project is essential for transforming AlphaCom's approach from reactive to proactive customer management.

Key Needs:

- Early Identification of at Risk Customers and Data Driven Decision Making
- Optimization of Retention Strategies and Revenue Protection
- Customer Experience Enhancement

Without this study, AlphaCom risks:

- Continued revenue decline and Increased customer acquisition costs

3. Understanding Business & Social Opportunities

Business Opportunities

- Targeted Marketing Campaigns and Improved Customer Lifetime Value (CLV)
- Product & Service Optimization and Competitive Advantage

Social Opportunities

- Better Customer Satisfaction

2. Exploratory Data Analysis

1. Data Collection and Background

- **Context:** AlphaCom, a telecommunications provider, is facing a significant increase in customer churn. This trend negatively impacts revenue and brand reputation in a highly competitive market.
- **Data Source:** The analysis is based on the customer_churn.csv dataset, which contains comprehensive information on customer demographics, service usage, billing, and contracts.

2. Data Overview

- **Dataset Dimensions**: The dataset consists of 12,055 rows and 20 columns.
- **Demographics**: Gender, SeniorCitizen, Partner, Dependents.
- **Services**: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Account Information**: Tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges.
- **Target Variable**: Churn

3. Univariate Analysis

1. Distribution of 'TenureGroup'

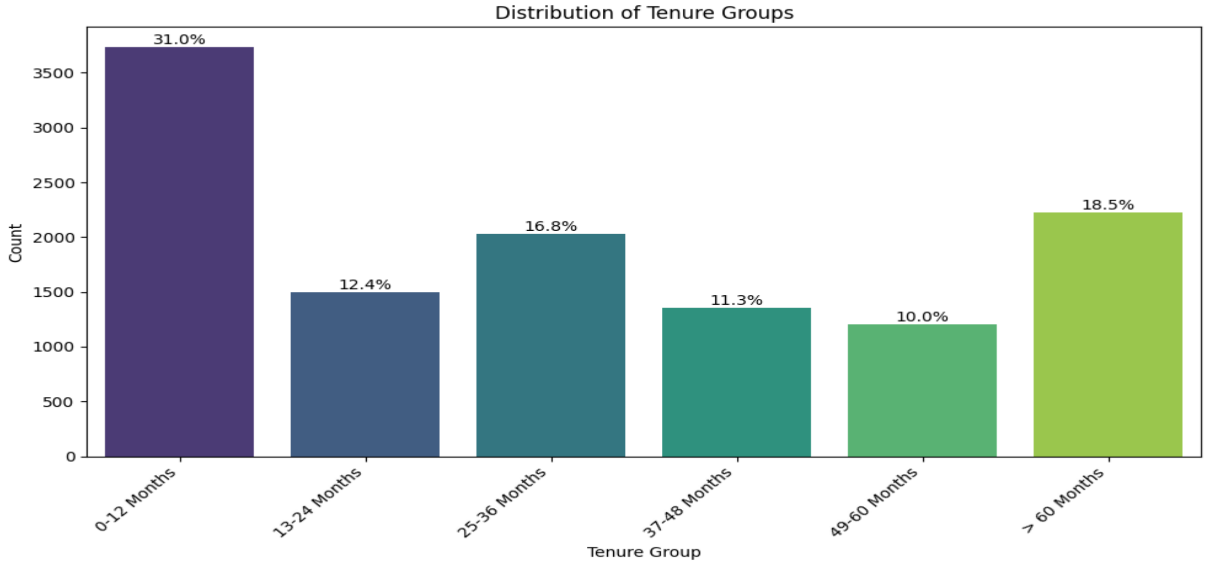


Fig. 1: Distribution of ' TenureGroup '

- The distribution of 'TenureGroup' shows that a large proportion of customers are in the '0-12 Months' group, indicating a significant number of new customers.

2. Distribution of 'HasInternet'

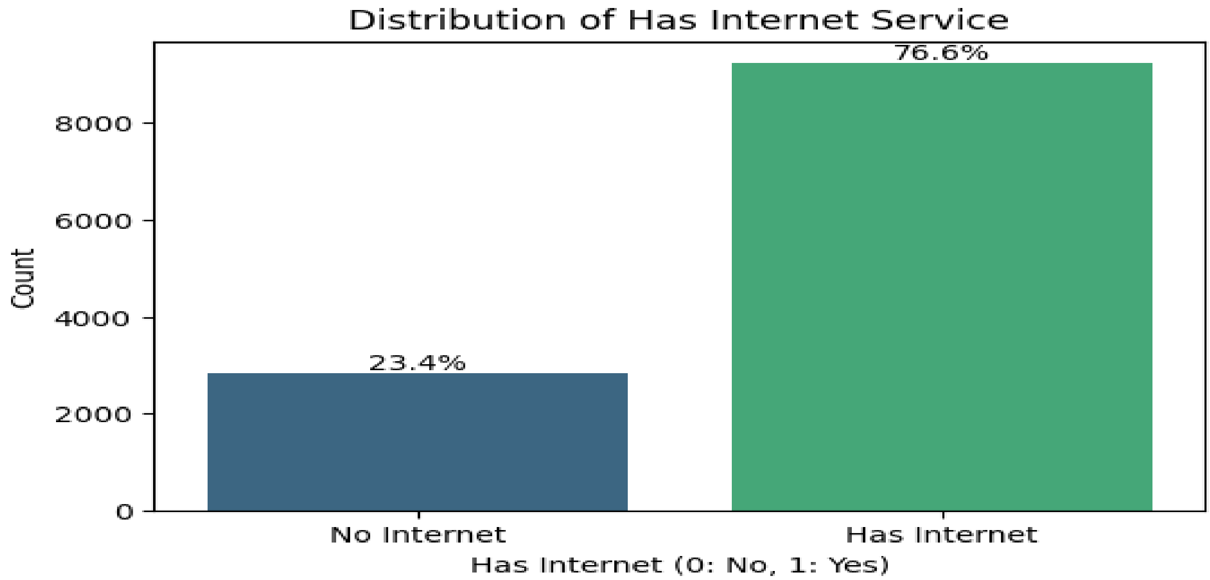


Fig. 2: Distribution of ' HasInternet '

- The majority of customers (76.6%) have internet service, while 23.4% do not.

3. Distribution of 'HasSecurity'

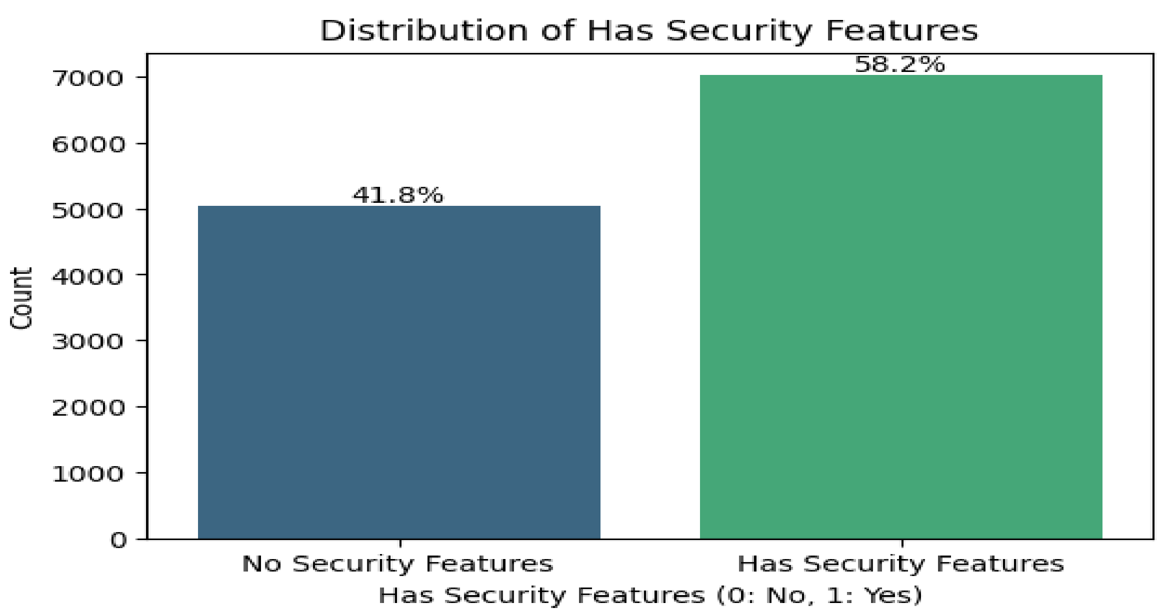


Fig. 3: Distribution of ' HasSecurity '

- Approximately 41.80% of customers do not have any of these security-related features, while 58.2% have at least one.

4. Distribution of 'TotalServices'

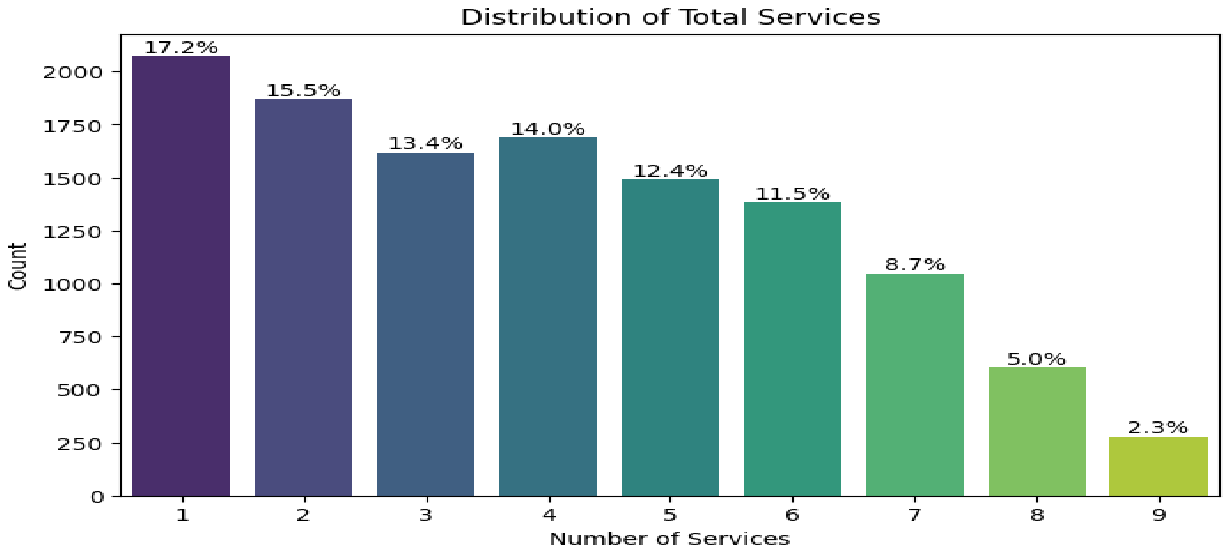


Fig. 4: Distribution of ' TotalServices '

- The distribution shows that many customers have a low number of services (around 1-3), while there's also a considerable number of customers with a higher number of services (around 6-7). This indicates a diverse customer base in terms of service adoption.

4. Bivariate Analysis

1. Churn by TenureGroup

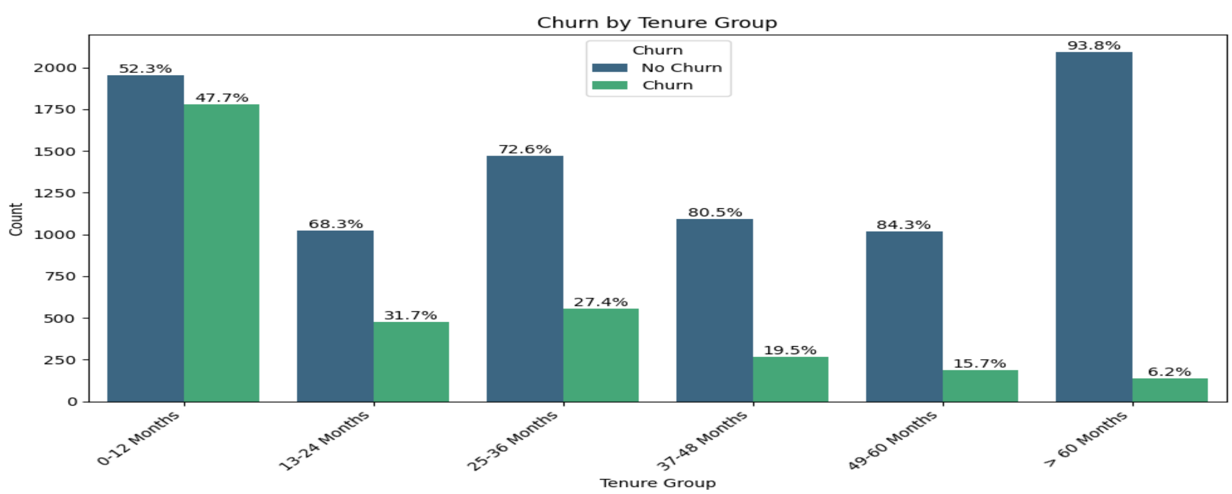


Fig. 5: Churn by TenureGroup

- **'0-12 Months' Group:** This group has the highest churn rate, significantly higher than all other tenure groups.

- **Higher Tenure Groups:** The churn rate generally decreases as tenure increases. Customers in the '> 60 Months' group have the lowest churn rate, indicating strong loyalty.

2. **Churn by HasInternet**

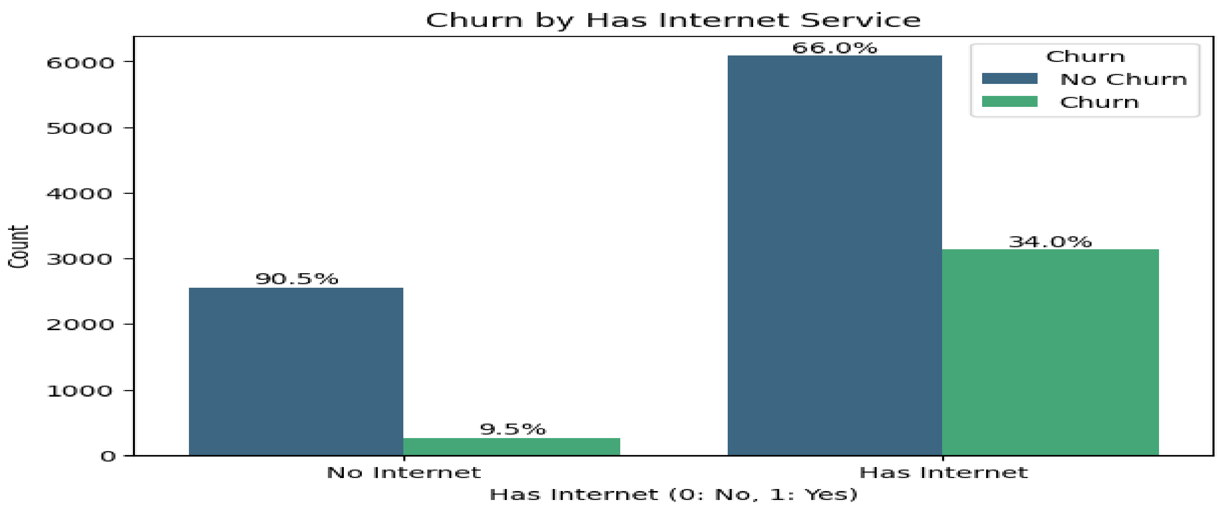


Fig. 6: Churn by HasInternet

- Customers who do not have internet service ('No Internet') have a very low churn rate of approximately 9.5%.
- Customers who do have internet service ('Has Internet') have a significantly higher churn rate of approximately 34.0%.

3. **Churn by HasSecurity**

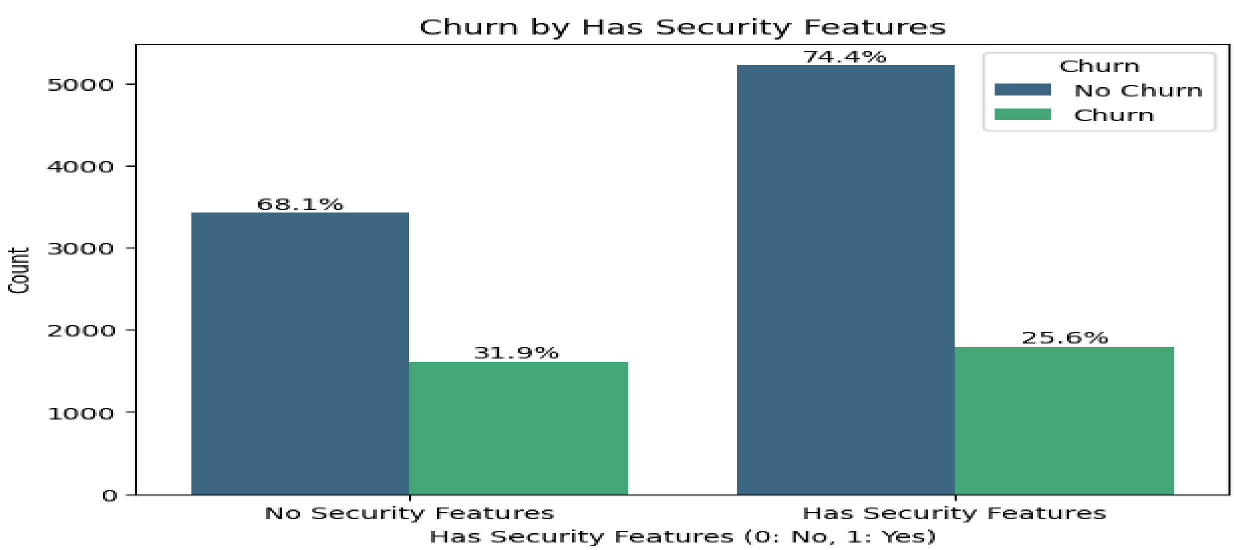


Fig. 7: Churn by HasSecurity

- Customers who do not have any security features have a significantly higher churn rate of approximately 31.9%.
- Customers who have at least one security feature have lower churn rate of approximately 25.6%.

4. Churn by TotalServices

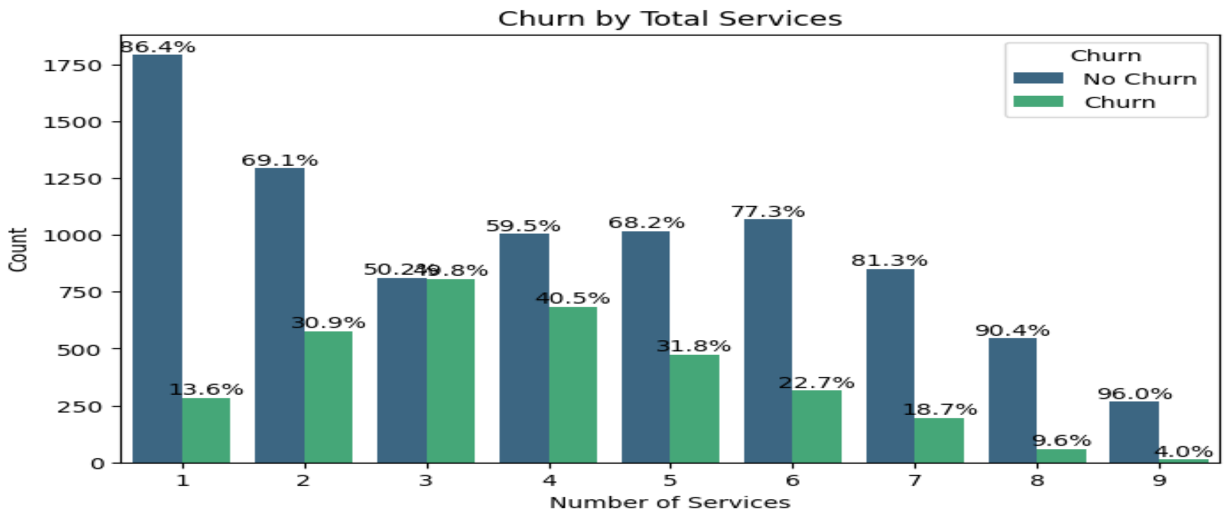


Fig. 8: Churn by TotalServices

- The churn rate appears to be highest for customers with a moderate number of services (around 2-3 services).
- Customers with only one service have a relatively low churn rate, possibly aligning with those who have 'No internet service'.
- Customers with a high number of services (6-9) also show a lower churn rate, suggesting that customers who are more deeply embedded with the company's offerings are more loyal.

3. Data Preprocessing

1. Duplicate value check and treatment

We identified four duplicate records within the dataset and removed them to ensure data integrity.

2. Data Cleaning

- **Target Variable:** The Churn column initially contained inconsistent strings (e.g., ' No ', 'NO', 'YES', 'yes', 'Yes', 'No', 'no', ' Yes '). These were standardized and converted into 'yes' and 'no'.
- **Financial Feature Correction:** MonthlyCharges and TotalCharges were imported as text due to currency symbols ('$' and '£'). We removed these symbols and converted the columns to float types.
- **Payment Method:** Leading and trailing spaces were stripped from categorical features like PaymentMethod to ensure unique categories were not duplicated by formatting errors.

3. Anomalous value check and treatment

- **Negative Value Treatment:** We identified impossible negative values in tenure and TotalCharges. These were treated as data entry errors and replaced with the median value of their respective columns to maintain data distribution without introducing bias.
- **Financial Feature:** Almost all rows (11,871) use dollars, while only a few (183) use pounds. This is likely a typing error, so we changed everything to dollars to keep the data consistent.

4. Missing value check and treatment

- **Missing Value Imputation:** We found 301 missing values in MonthlyCharges and 1,205 in TotalCharges. We filled these values using the median value because it is more reliable than the average when dealing with outliers.

5. Outlier check and treatment

- After checking all numerical columns, only TotalCharges showed outliers. The data is right-skewed, meaning several customers have much higher charges than the rest. However, we decided to keep these points because they represent long-term, high-value customers. Since these are real, valid records, removing them would lose important information.

6. Feature engineering

- **Categorical Encoding:** Since machine learning models require numerical input, categorical variables were converted using One-Hot Encoding.

7. Feature scaling

- To ensure features like TotalCharges (values in thousands) do not mathematically overpower SeniorCitizen (0 or 1), we applied feature scaling (e.g., StandardScalar) to bring all numeric features to a comparable range.

8. Data preparation for modeling

We completed the data preprocessing by scaling the features and splitting the dataset into an 70/15/15 i.e Train/Validate/Test split. This ensures the model is validated on unseen data to prevent overfitting. The resulting sets are:

- Training Set: 8435 samples
- Validation Set: 1808 samples
- Testing Set: 1808 samples

9. Data leakage handling

To ensure the integrity of our results, we isolated the training, validation, and test sets. After training the model on the training dataset, we utilized the validation set to fine-tune hyperparameters and compare different architectures.

Only after selecting the best performing model based on these validation results did we introduce the test set.

4. Model Building - Baseline Model

1. Metric of Choice

The target variable is binary (churn or no churn) and the dataset is imbalanced (fewer churned customers than non-churned), specific evaluation metrics are more appropriate than simple accuracy.

Here are the chosen metrics with their rationale:

1. **Recall for the 'Churn' class :**

- Rationale: Recall measures the proportion of actual churners that were correctly identified by the model. (minimize False Negatives) Missing a churner means losing a customer, which directly impacts revenue.

2. **Precision for the 'Churn' class:**

- Rationale: Precision measures the proportion of customers predicted as churners who actually churned.(minimize False Positives). High precision ensures that retention efforts are targeted effectively and resources are not wasted on customers who wouldn't churn anyway.

3. **F1-Score:**

- Rationale: The F1-Score is the harmonic mean of Precision and Recall. A high F1-score indicates a good balance between identifying positive cases and not making too many false positive predictions.

4. **ROC AUC (Receiver Operating Characteristic - Area Under the Curve):**

- Rationale: ROC AUC measures the ability of the model to distinguish between positive and negative classes across various classification thresholds. A higher AUC indicates a better model performance in separating churners from non-churners.

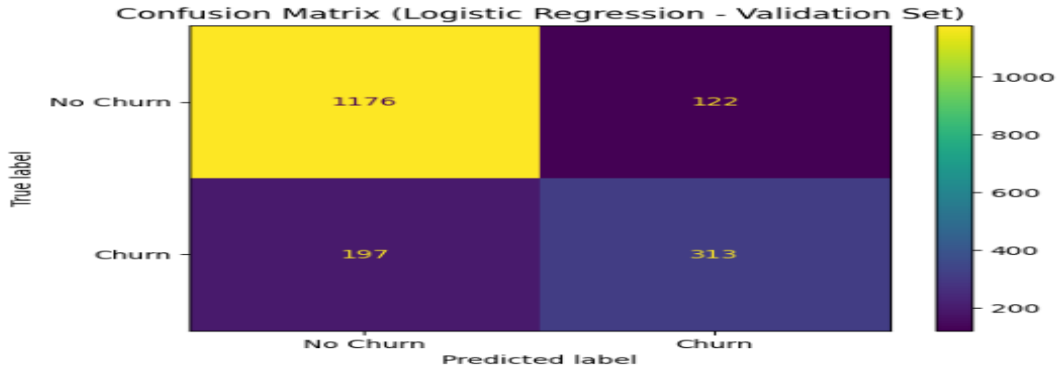
In summary, while we will monitor accuracy, **the primary focus will be on Recall, Precision, and F1-Score for the 'Churn' class**, and **ROC AUC for overall model performance**, to ensure that the model is effective in identifying and acting upon potential customer churn.

In a churn scenario, the cost of **False Negatives** (failing to identify a customer who is about to leave) is much higher than the cost of **False Positives** (offering a discount to someone who was going to stay anyway). The main choice will be **Recall**.

2. Build a baseline model

1. Logistic Regression

```
Confusion Matrix (Logistic Regression - Validation Set):
```



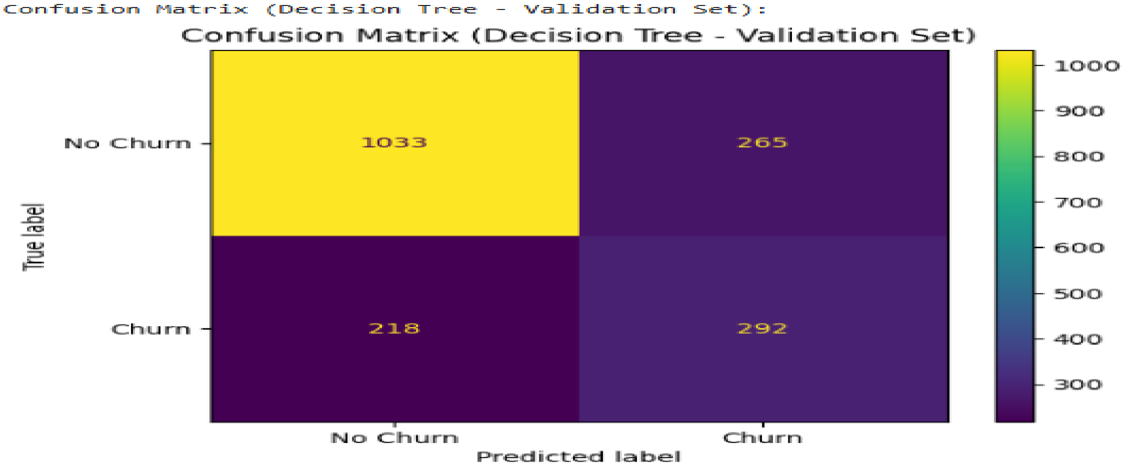

```
Classification Report (Logistic Regression - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.86      0.91      0.88      1298
       Churn       0.72      0.61      0.66       510

    accuracy                           0.82      1808
   macro avg       0.79      0.76      0.77      1808
weighted avg       0.82      0.82      0.82      1808

ROC AUC Score (Logistic Regression - Validation Set): 0.8638
```

Fig. 9: Logistic Regression

5. Model Building - Advanced Model

1. Decision Tree Classifier

```
Confusion Matrix (Decision Tree - Validation Set):
```

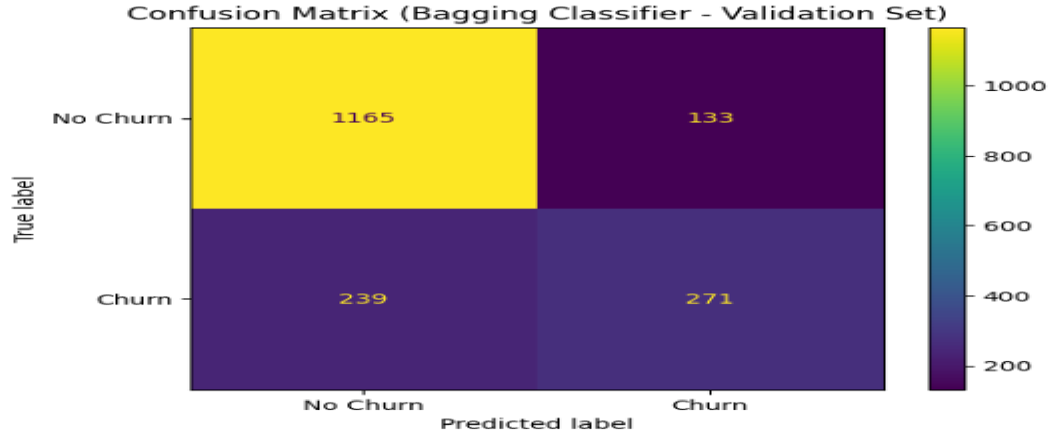



```
Classification Report (Decision Tree - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.83      0.80      0.81      1298
       Churn       0.52      0.57      0.55       510

    accuracy                           0.73      1808
   macro avg       0.67      0.68      0.68      1808
weighted avg       0.74      0.73      0.74      1808

ROC AUC Score (Decision Tree - Validation Set): 0.6842
```

Fig. 10: Decision Tree Classifier

2. Bagging Classifier

Confusion Matrix (Bagging Classifier - Validation Set):

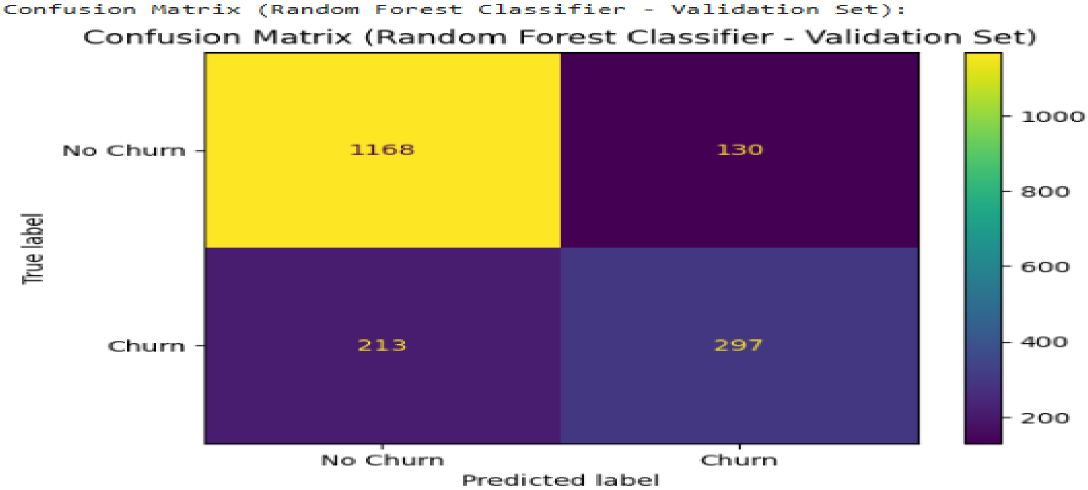



```
Classification Report (Bagging Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.83      0.90      0.86      1298
       Churn       0.67      0.53      0.59       510

    accuracy                           0.79      1808
   macro avg       0.75      0.71      0.73      1808
weighted avg       0.78      0.79      0.79      1808

ROC AUC Score (Bagging Classifier - Validation Set): 0.8232
```

Fig. 11: Bagging Classifier

3. Random Forest Classifier

Confusion Matrix (Random Forest Classifier - Validation Set):

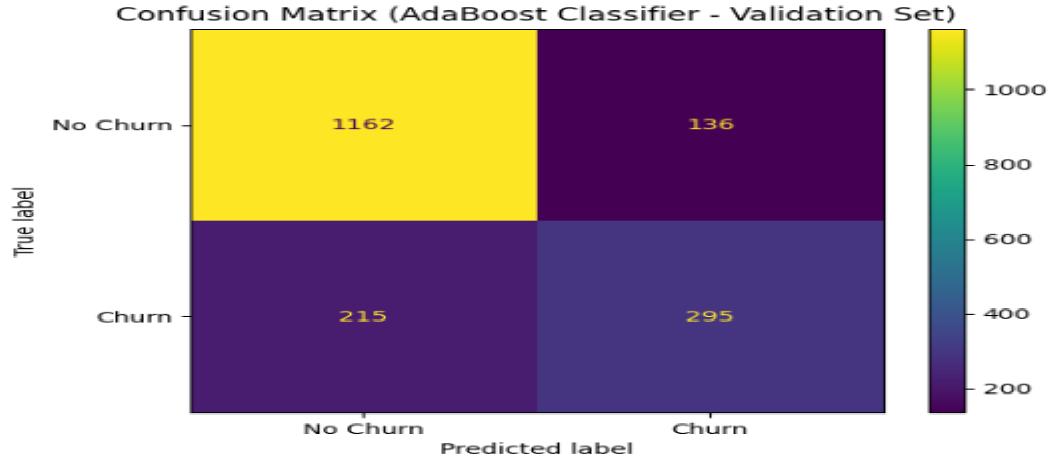



```
Classification Report (Random Forest Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.90      0.87      1298
       Churn       0.70      0.58      0.63       510

    accuracy                           0.81      1808
   macro avg       0.77      0.74      0.75      1808
weighted avg       0.80      0.81      0.80      1808

ROC AUC Score (Random Forest Classifier - Validation Set): 0.8479
```

Fig. 12: Random Forest Classifier

4. AdaBoost Classifier

```
Confusion Matrix (AdaBoost Classifier - Validation Set):
```

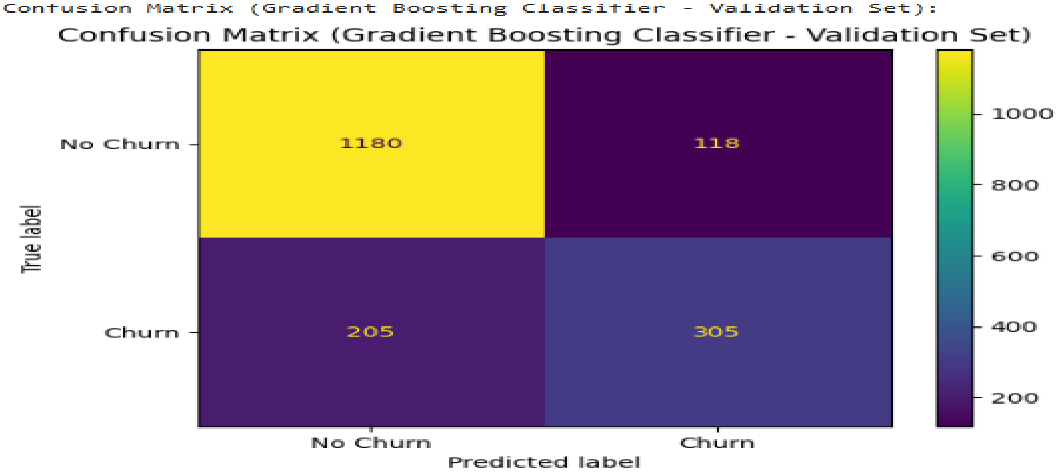



```
Classification Report (AdaBoost Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.84      0.90      0.87      1298
       Churn       0.68      0.58      0.63       510

    accuracy                           0.81      1808
   macro avg       0.76      0.74      0.75      1808
weighted avg       0.80      0.81      0.80      1808

ROC AUC Score (AdaBoost Classifier - Validation Set): 0.8569
```

Fig. 13: AdaBoost Classifier

5. Gradient Boosting Classifier

```
Confusion Matrix (Gradient Boosting Classifier - Validation Set):
```

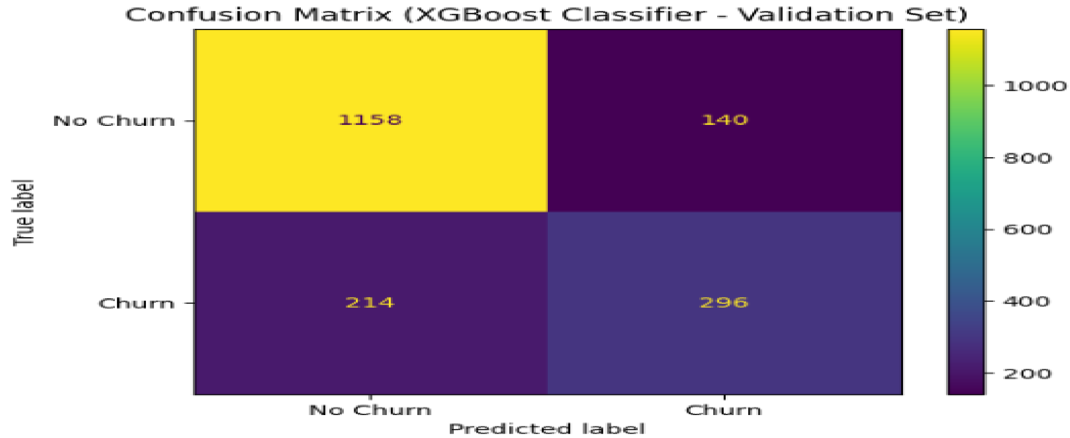



```
Classification Report (Gradient Boosting Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.91      0.88      1298
       Churn       0.72      0.60      0.65       510

    accuracy                           0.82      1808
   macro avg       0.79      0.75      0.77      1808
weighted avg       0.82      0.82      0.82      1808

ROC AUC Score (Gradient Boosting Classifier - Validation Set): 0.8647
```

Fig. 14: Gradient Boosting Classifier

6. XGBoost Classifier

```
Confusion Matrix (XGBoost Classifier - Validation Set):
```




```
Classification Report (XGBoost Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.84      0.89      0.87      1298
       Churn       0.68      0.58      0.63       510

    accuracy                           0.80      1808
   macro avg       0.76      0.74      0.75      1808
weighted avg       0.80      0.80      0.80      1808

ROC AUC Score (XGBoost Classifier - Validation Set): 0.8509
```

Fig. 15: XGBoost Classifier

Model Performance Comparison

Let's compare the performance of all the models on the validation set using our chosen evaluation metrics.

	Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)	ROC AUC
6	Gradient Boosting Classifier	0.8213	0.7210	0.5980	0.6538	0.8647
0	Logistic Regression	0.8236	0.7195	0.6137	0.6624	0.8638
5	AdaBoost Classifier	0.8059	0.6845	0.5784	0.6270	0.8569
7	XGBoost Classifier	0.8042	0.6789	0.5804	0.6258	0.8509
4	Random Forest Classifier	0.8103	0.6956	0.5824	0.6339	0.8479
3	Bagging Classifier	0.7942	0.6708	0.5314	0.5930	0.8232
2	K-Nearest Neighbors	0.7572	0.5697	0.5686	0.5692	0.7875
1	Decision Tree	0.7329	0.5242	0.5725	0.5473	0.6842

Table 1: Model Performance Comparison

Observations

Based on the evaluation metrics, particularly focusing on Recall (Churn), Precision (Churn), F1-Score (Churn), and ROC AUC, we can make the following observations:

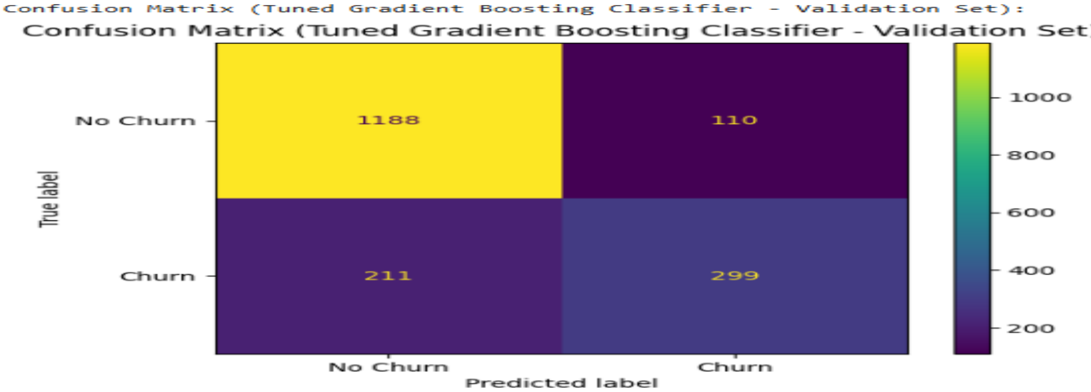
- **Overall Best Performers:**
    - Logistic Regression and Gradient Boosting Classifier show the best overall performance with the highest ROC AUC scores (0.8638 and 0.8647 respectively). These models also exhibit a good balance across precision and recall for the churn class.
- **Recall (Churn):**
    - Logistic Regression (0.61) and Gradient Boosting Classifier (0.60), AdaBoost Classifier (0.58) and XGBoost Classifier (0.58), and Random Forest Classifier (0.58) achieve relatively good recall scores, meaning they are better at identifying actual churners. This is crucial for AlphaCom to proactively target at risk customers.
- **Precision (Churn):**
    - Logistic Regression (0.72) and Gradient Boosting Classifier (0.72) demonstrate the highest precision for the churn class. This means that when these models predict a customer will churn, they are more likely to be correct, leading to more efficient resource allocation for retention campaigns.
    - AdaBoost Classifier (0.68), XGBoost Classifier (0.68), Bagging Classifier (0.67), and Random Forest Classifier (0.70) also show decent precision scores.
    - Decision Tree (0.52) and K-Nearest Neighbors (0.57) have the lowest precision, indicating a higher rate of false positives, which could lead to wasted retention efforts.
- **F1-Score (Churn):**
    - Logistic Regression (0.66) and Gradient Boosting Classifier (0.65) lead in F1-score, suggesting a strong balance between identifying churners and doing so accurately. This is a robust indicator for overall effectiveness in churn prediction.
- **ROC AUC:**
    - Gradient Boosting Classifier (0.8647) and Logistic Regression (0.8638) have the highest ROC AUC scores, indicating their superior ability to distinguish between churners and non churners across various classification thresholds. This makes them highly reliable for ranking customers by their churn risk.

6. Model Performance Improvement using Hyperparameter Tuning

From the previous obervations we will choose Gradient Boosting Classifier, Logistic Regression, AdaBoost Classifier, XGBoost Classifier and Random Forest Classifier for tunning.

1. Tuned Gradient Boosting Classifier

```
Confusion Matrix (Tuned Gradient Boosting Classifier - Validation Set):
```

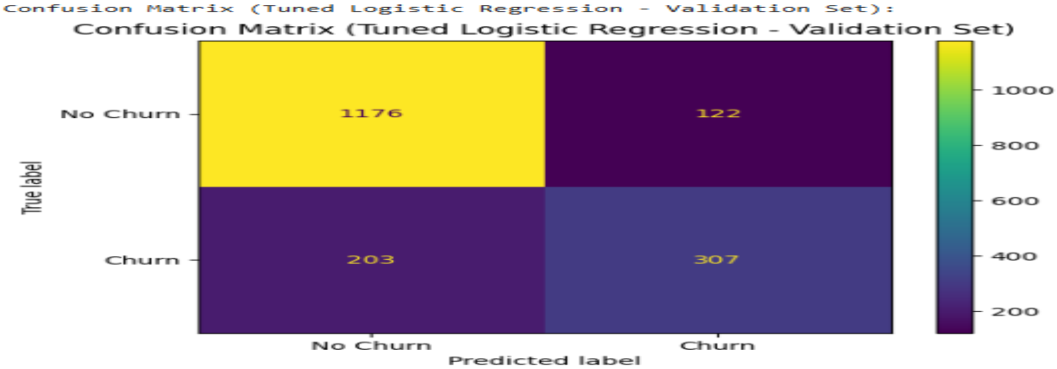



```
Classification Report (Tuned Gradient Boosting Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.92      0.88      1298
       Churn       0.73      0.59      0.65       510

    accuracy                           0.82      1808
   macro avg       0.79      0.75      0.77      1808
weighted avg       0.82      0.82      0.82      1808

ROC AUC Score (Tuned Gradient Boosting Classifier - Validation Set): 0.8634
```

Fig. 16: Tuned Gradient Boosting Classifier

2. Tuned Logistic Regression

```
Confusion Matrix (Tuned Logistic Regression - Validation Set):
```

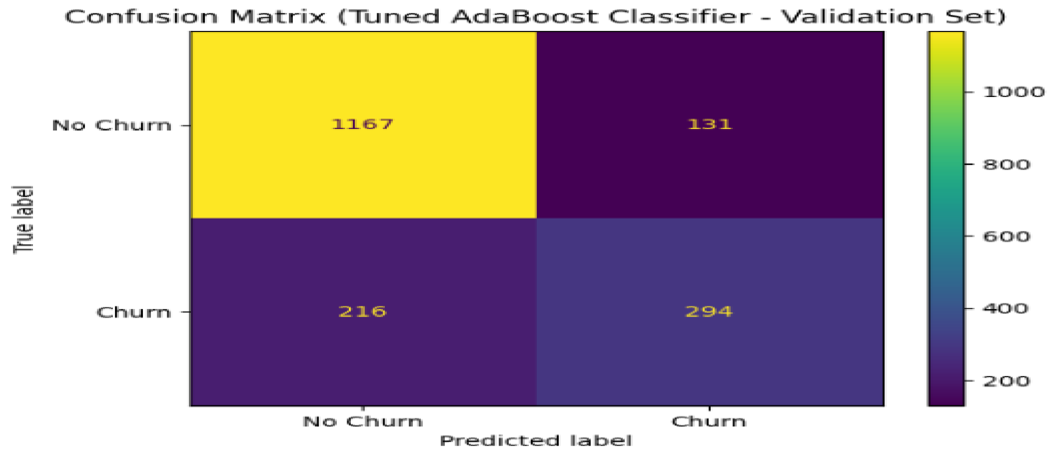



```
Classification Report (Tuned Logistic Regression - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.91      0.88      1298
       Churn       0.72      0.60      0.65       510

    accuracy                           0.82      1808
   macro avg       0.78      0.75      0.77      1808
weighted avg       0.81      0.82      0.82      1808

ROC AUC Score (Tuned Logistic Regression - Validation Set): 0.8618
```

Fig. 17: Tuned Logistic Regression

3. Tuned AdaBoost Classifier

```
Confusion Matrix (Tuned AdaBoost Classifier - Validation Set):
```

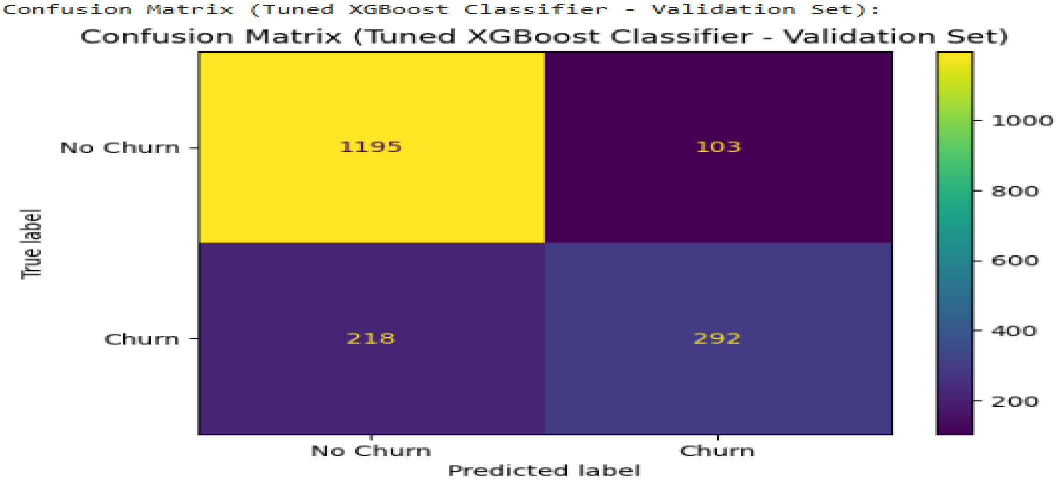



```
Classification Report (Tuned AdaBoost Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.84      0.90      0.87      1298
       Churn       0.69      0.58      0.63       510

    accuracy                           0.81      1808
   macro avg       0.77      0.74      0.75      1808
weighted avg       0.80      0.81      0.80      1808

ROC AUC Score (Tuned AdaBoost Classifier - Validation Set): 0.8567
```

Fig. 18: Tuned AdaBoost Classifier

4. Tuned XGBoost Classifier

```
Confusion Matrix (Tuned XGBoost Classifier - Validation Set):
```




```
Classification Report (Tuned XGBoost Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.92      0.88      1298
       Churn       0.74      0.57      0.65       510

    accuracy                           0.82      1808
   macro avg       0.79      0.75      0.76      1808
weighted avg       0.82      0.82      0.81      1808

ROC AUC Score (Tuned XGBoost Classifier - Validation Set): 0.8620
```

Fig. 19: Tuned XGBoost Classifier

Tuned Model Performance Comparison

Let's compare the performance of all the tuned models on the validation set using our chosen evaluation metrics.

	Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)	ROC AUC
0	Tuned Gradient Boosting Classifier	0.8225	0.7311	0.5863	0.6507	0.8634
3	Tuned XGBoost Classifier	0.8225	0.7392	0.5725	0.6453	0.8620
1	Tuned Logistic Regression	0.8202	0.7156	0.6020	0.6539	0.8618
4	Tuned Random Forest Classifier	0.8180	0.7191	0.5824	0.6436	0.8604
2	Tuned AdaBoost Classifier	0.8081	0.6918	0.5765	0.6289	0.8567

Table 2: Tuned Model Performance Comparison

Observations

Based on the evaluation metrics, particularly focusing on Recall (Churn), Precision (Churn), F1-Score (Churn), and ROC AUC, we can make the following observations:

- **Overall Best Performers:**
    - Tuned XGBoost and Tuned Gradient Boosting Classifier show the best overall performance with the highest ROC AUC scores (0.8620 and 0.8634 respectively). These models also exhibit a good balance across precision and recall for the churn class.
- **Recall (Churn):**
    - Tuned Logistic Regression (0.60) and Tuned Gradient Boosting Classifier (0.59), Tuned Random Forest Classifier (0.58) and Tuned AdaBoost Classifier (0.58), and Tuned Classifier (0.57) achieve relatively good recall scores, meaning they are better at identifying actual churners.
- **Precision (Churn):**
    - Tuned XGBoost Classifier (0.74) and Tuned Gradient Boosting Classifier (0.73) demonstrate the highest precision for the churn class.
    - Tuned Logistic Regression (0.72), Tuned Random Forest Classifier (0.72) and Tuned AdaBoost Classifier (0.69) also show decent precision scores.

- **F1-Score (Churn):**
  - Tuned Gradient Boosting Classifier (0.65) and Tuned XGBoost Classifier (0.65) lead in F1-score, suggesting a strong balance between identifying churners and doing so accurately. This is a robust indicator for overall effectiveness in churn prediction.
- **ROC AUC:**
  - Tuned Gradient Boosting Classifier (0.8634) and Tuned XGBoost Classifier (0.8620) have the highest ROC AUC scores, indicating their superior ability to distinguish between churners and non churners across various classification thresholds.

7. Model Performance Comparison and Final Model Selection

Let's compare performance of all the untuned and tuned models,

	Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)	ROC AUC
0	Gradient Boosting Classifier	0.8213	0.7210	0.5980	0.6538	0.8647
1	Logistic Regression	0.8236	0.7195	0.6137	0.6624	0.8638
2	Tuned Gradient Boosting Classifier	0.8225	0.7311	0.5863	0.6507	0.8634
3	Tuned XGBoost Classifier	0.8225	0.7392	0.5725	0.6453	0.8620
4	Tuned Logistic Regression	0.8202	0.7156	0.6020	0.6539	0.8618
5	Tuned Random Forest Classifier	0.8180	0.7191	0.5824	0.6436	0.8604
6	AdaBoost Classifier	0.8059	0.6845	0.5784	0.6270	0.8569
7	Tuned AdaBoost Classifier	0.8081	0.6918	0.5765	0.6289	0.8567
8	XGBoost Classifier	0.8042	0.6789	0.5804	0.6258	0.8509
9	Random Forest Classifier	0.8103	0.6956	0.5824	0.6339	0.8479
10	Bagging Classifier	0.7942	0.6708	0.5314	0.5930	0.8232
11	K-Nearest Neighbors	0.7572	0.5697	0.5686	0.5692	0.7875
12	Decision Tree	0.7329	0.5242	0.5725	0.5473	0.6842

Table 3: Untuned and Tuned Model Performance Comparison

Overall Best Model

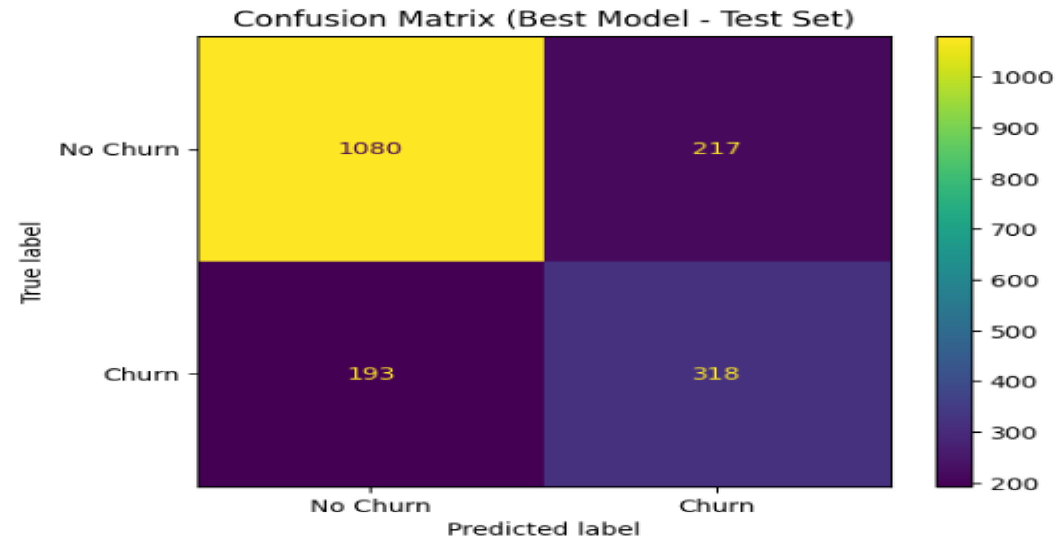
Based on the combined results of both untuned and tuned models, sorted by ROC AUC:

- The untuned Gradient Boosting Classifier stands out as the best performing model with the highest ROC AUC score of 0.8647.
- It also demonstrates a strong balance across other critical metrics for the 'Churn' class:
  - Precision (Churn): 0.7200
  - Recall (Churn): 0.6000
  - F1-Score (Churn): 0.6500

Although hyperparameter tuning was performed, in this specific instance, the default parameters of the Gradient Boosting Classifier yielded slightly better results on the validation set than its tuned counterpart. The **untuned Gradient Boosting Classifier** is the best model for this churn prediction task.

Best Model Performance On Test Set

```
Confusion Matrix (Best Model - Test Set):
```




```
Classification Report (Best Model - Test Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.83      0.84      1297
       Churn       0.59      0.62      0.61       511

    accuracy                           0.77      1808
   macro avg       0.72      0.73      0.72      1808
weighted avg       0.78      0.77      0.77      1808

ROC AUC Score (Best Model - Test Set): 0.8056
```

Fig. 20: Best Model Performance On Test Set

Observations:

The evaluation of the **untuned Gradient Boosting Classifier** on the test set reveals the following:

- **Accuracy:** The model achieved an accuracy of approximately **0.77**, indicating it correctly classified about 77% of the customers in the test set.
- **Precision (Churn):** The precision for the 'Churn' class is around **0.59**. This means that when the model predicts a customer will churn, it is correct approximately 59% of the time.
- **Recall (Churn):** The recall for the 'Churn' class is approximately **0.62**. This indicates that the model successfully identified about 62% of the actual churners in the test set.
- **F1-Score (Churn):** The F1-score for the 'Churn' class is approximately **0.61**. This score provides a balanced measure of precision and recall, reflecting a solid performance in identifying churners.

- **ROC AUC Score:** The ROC AUC score on the test set is approximately **0.8056.** This value signifies that the model has a strong ability to distinguish between churners and non-churners across various classification thresholds.

Overall, the model's ability to identify actual churners (recall) is strong, which is a key metric for this task.

8. Actionable Insights & Recommendations

Key Insights from Analysis

1. Contract Type Matters: Month-to-month contracts are associated with the highest churn rate (approx. 46%), while longer-term contracts (one-year and two-year) drastically reduce churn (11.3% and 3.3% respectively).
2. Early Churn is Critical: A large proportion of churn occurs within the first 12 months of service, indicating a potential issue with initial customer experience or onboarding.
3. Fiber Optic Service and Churn: Customers using Fiber Optic internet service exhibit a significantly higher churn rate (approx. 44.9%) compared to DSL users or those without internet service. This suggests potential dissatisfaction with Fiber Optic service quality, pricing, or support.
4. Payment Method Impact: Customers using electronic checks have the highest churn rate (approx. 49.1%), whereas automatic payment methods (bank transfer and credit card) are linked to much lower churn.
5. Internet Security Features Drive Retention: The absence of online security, online backup, device protection, or tech support features correlates with considerably higher churn rates. Customers who opt for these features are notably more loyal.

Actionable Business Recommendations

1. Strategize for Longer-Term Contracts: Actively encourage customers to switch from month-to-month to one-year or two-year contracts by offering significant incentives such as reduced monthly rates, free upgrades, or waived fees for committing to longer terms.
2. Prioritize New Customer Retention Programs: Implement targeted onboarding programs and proactive outreach for new customers (especially within their first year) to address potential issues early and foster engagement through introductory discounts or dedicated support channels.
3. Investigate and Improve Fiber Optic Service: Conduct in-depth analysis into the root causes of high churn among Fiber Optic users. Survey these customers, review service outage reports, and consider service enhancements or competitive pricing adjustments.
4. Optimize Payment Method Options: Analyze issues associated with electronic checks and promote automatic payment methods. Incentivize customers (e.g., with small

monthly discounts) to switch to automatic bank transfers or credit card payments, emphasizing convenience.

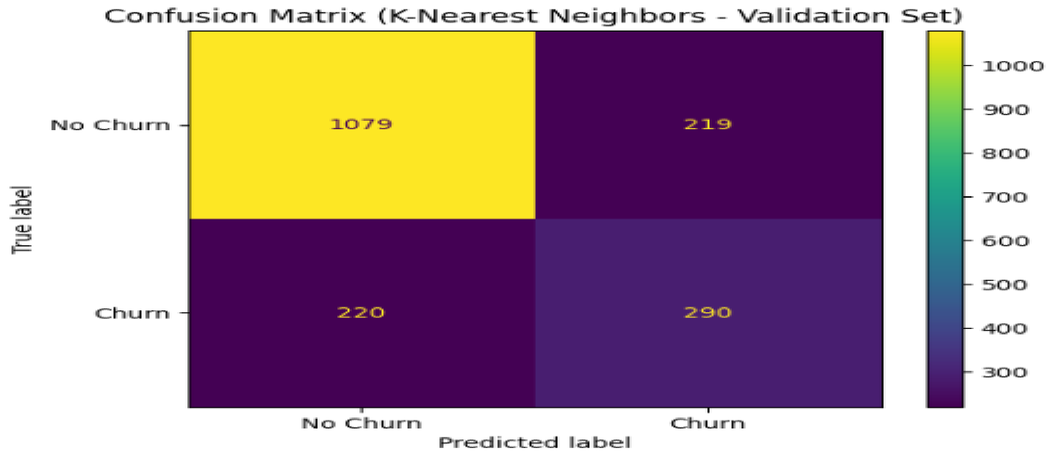
5. Promote and Incentivize Security and Support Services: Highlight the benefits of Online Security, Online Backup, Device Protection, and Tech Support. Bundle these services into attractive packages, offer free trials, or provide discounts for adoption to reduce churn and increase customer stickiness.

Thank You.

Appendix:

5.7. K-Nearest Neighbors Classifier

```
Confusion Matrix (K-Nearest Neighbors - Validation Set):
```

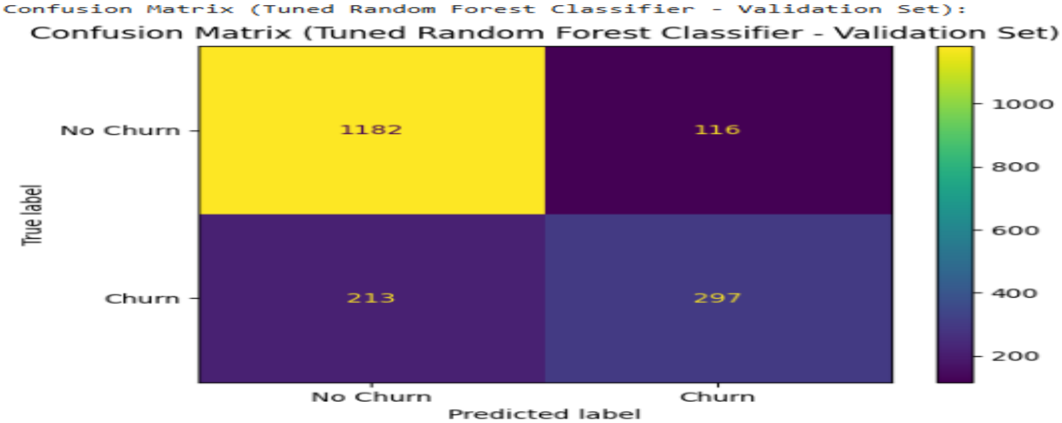



```
Classification Report (K-Nearest Neighbors - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.83      0.83      0.83      1298
       Churn       0.57      0.57      0.57       510

    accuracy                           0.76      1808
   macro avg       0.70      0.70      0.70      1808
weighted avg       0.76      0.76      0.76      1808

ROC AUC Score (K-Nearest Neighbors - Validation Set): 0.7875
```

Fig 5.7: K-Nearest Neighbors Classifier

6.5. Tuned Random Forest Classifier

```
Confusion Matrix (Tuned Random Forest Classifier - Validation Set):
```




```
Classification Report (Tuned Random Forest Classifier - Validation Set):
              precision    recall  f1-score   support

    No Churn       0.85      0.91      0.88      1298
       Churn       0.72      0.58      0.64       510

    accuracy                           0.82      1808
   macro avg       0.78      0.75      0.76      1808
weighted avg       0.81      0.82      0.81      1808

ROC AUC Score (Tuned Random Forest Classifier - Validation Set): 0.8604
```

Fig 6.5: Tuned Random Forest Classifier